

Tool-98 — Operational Loss Calculator

Capstone Tool | Team: Team: 5 Members

Sprint: Monday 14 April 2026 — Friday 9 May 2026 | Demo Day: 9 May 2026

[!] **READ THIS DOCUMENT COMPLETELY BEFORE WRITING A SINGLE LINE OF CODE.** This is your complete project guide — what to build, the full tech stack, exact folder structure, every task broken down day by day, GitHub submission steps, Demo Day checklist, and the 8 most common mistakes. Read it twice.

1. PROJECT OVERVIEW

You are building **Tool-98 — Operational Loss Calculator** as your internship capstone project. This is a real-world, AI-powered web application built using an industry-standard tech stack. It will be the strongest item on your resume when you graduate.

Tool ID	Tool-98
Tool Name	Operational Loss Calculator
Tool Type	Capstone — Focused Tool
Team	Team: 5 Members
Sprint	Monday 14 April – Friday 9 May 2026 (20 working days)
Demo Day	Friday 9 May 2026 — 6-minute live presentation
Backend Port	8080 Swagger: http://localhost:8080/swagger-ui.html
AI Port	5000 Health: http://localhost:5000/health
Frontend Port	80 Access: http://localhost
Repository	Link will be shared by your mentor before Day 1

2. TECH STACK

Use this exact stack. Do not change it. Before Day 2 read the getting-started docs for any technology you are unfamiliar with: spring.io/guides, console.groq.com/docs, docs.trychroma.com, react.dev, tailwindcss.com, docs.docker.com/compose.

Technology	What It Does in This Project
Java 17	Backend language. Download from adoptium.net
Spring Boot 3.x	Java framework — REST APIs, security, scheduling
PostgreSQL 15	Primary database. Run inside Docker
Redis 7	In-memory cache and AI response cache
Flyway	DB migrations — every schema change in a numbered SQL file
Spring Security + JWT	Login and role-based access control
Python 3.11	AI service language
Flask 3.x	Python web framework — AI microservice on port 5000
Groq API (LLaMA-3.3-70b)	AI model — free tier at console.groq.com , no credit card
flask-limiter	Rate limiting — blocks IPs exceeding 30 req/min
React 18 + Vite	Frontend framework with fast build tool
Tailwind CSS	Utility CSS — style with class names
Axios	HTTP client for React to call the Java backend
Docker + Docker Compose	Runs all services with one command: <code>docker-compose up</code>

3. FOLDER STRUCTURE — Create This on Day 1

Create all folders on Day 1 before writing any code. Use this exact layout throughout the sprint.

```
your-project-folder/
|-- backend/ <- Spring Boot project
| |-- src/main/java/com/internship/tool/
| | |-- controller/ <- REST endpoints
| | |-- service/ <- Business logic
| | |-- repository/ <- DB queries
| | |-- entity/ <- JPA table models
| | |-- config/ <- Security, Redis, Mail
| | |-- exception/ <- Custom exceptions
| |-- src/main/resources/
| | |-- db/migration/ <- V1__init.sql, V2__...
| | |-- application.yml <- All config (env variable references)
| +-- pom.xml
|
|-- ai-service/ <- Flask Python microservice
| |-- routes/ <- endpoint files
| |-- services/ <- groq_client.py
| |-- prompts/ <- prompt template text files
| |-- app.py
| |-- Dockerfile
| +-- requirements.txt
|
|-- frontend/ <- React + Vite frontend
| |-- src/
| | |-- components/
| | |-- pages/
| | |-- services/
| | +-- App.jsx
| +-- package.json
|
|-- docker-compose.yml
|-- .env.example
+-- README.md
```

4. TEAM ROLES

Your mentor will coordinate teams and guide progress. Each role below has a clear set of responsibilities for the full sprint.

Role	Responsibilities
Java Developer 1	Spring Boot setup, Service layer, JWT auth, Docker Compose, data seeder, README.md, opens Demo Day.
Java Developer 2	DB schema (Flyway), Repository layer, full React frontend, email notifications, demo video.
AI Developer 1	Flask setup, prompt templates, /describe and /recommend endpoints, AIServiceClient.java.
AI Developer 2	GroqClient, /generate-report, security review, prompt tuning, AI docs, AI talking points.
Security Reviewer	Security testing, validates fixes, presents security features on Demo Day (if assigned).

Your Team

Member	Role	Contact
Member 1	Role will be confirmed by your mentor before Day 1	Contact your mentor for team WhatsApp group details
Member 2	Role will be confirmed by your mentor before Day 1	Contact your mentor for team WhatsApp group details
Member 3	Role will be confirmed by your mentor before Day 1	Contact your mentor for team WhatsApp group details
Member 4	Role will be confirmed by your mentor before Day 1	Contact your mentor for team WhatsApp group details
Member 5	Role will be confirmed by your mentor before Day 1	Contact your mentor for team WhatsApp group details

5. DAY-BY-DAY TASK BREAKDOWN (20 Working Days)

Each row is one task for one role on one specific day. Complete it fully before end of that working day. If blocked, tell your team immediately — do not stay stuck silently. Update your status daily: **Not Started** → **In Progress** → **Completed** → **Blocked**

[YEL] Week 1: Foundation		[ORN] Week 2: Core Features		[BLU] Week 3: Polish	[GRN] Week 4: Demo Prep	[PUR] Day 20: Demo Day
Day	Date	Track	Role	Task — Complete This Fully Today		Status
Day 1	Mon Apr 14	— Week 1 —				
		Java	Java Developer 1	Setup Spring Boot — folder structure, pom.xml, application.yml with \${ENV_VAR} placeholders for DB, Redis, mail, JWT		Not Started
		Java	Java Developer 2	Write Flyway V1 migration — core table with all required columns, correct types, indexes on key fields		Not Started
		AI	AI Developer 1	Setup Flask — routes/, services/, prompts/ folders, requirements.txt, app.py entry point		Not Started
		AI	AI Developer 2	Get Groq API key at console.groq.com, store in .env, write test_groq.py and confirm API call works		Not Started
Day 2	Tue Apr 15	— Week 1 —				
		Java	Java Developer 1	Create JPA Entity and Repository — all fields, @Column, @CreatedDate/@LastModifiedDate, custom query methods		Not Started
		Java	Java Developer 2	Setup React with Vite — install Axios and Tailwind, set API base URL, create pages/ components/ services/ folders		Not Started
		AI	AI Developer 1	Write primary prompt template — test with 5 real inputs, refine until all 5 outputs are consistently good		Not Started
		AI	AI Developer 2	Implement GroqClient — API call, JSON parsing, 3-retry with backoff, error logging. Create SECURITY.md with 5 threats		Not Started
Day 3	Wed Apr 16	— Week 1 —				
		Java	Java Developer 1	Implement Service layer — business logic, input validation, custom exception classes for each error type		Not Started
		Java	Java Developer 2	Build list page — table, Axios GET on mount, loading state, empty state. Write Flyway V2 migration for audit_log		Not Started
		AI	AI Developer 1	Build POST /describe — validate input, load prompt, call Groq, return structured JSON with generated_at		Not Started
		AI	AI Developer 2	Add input sanitisation middleware — strip HTML, detect prompt injection, return 400. Add flask-limiter 30 req/min		Not Started
Day 4	Thu Apr 17	— Week 1 —				
		Java	Java Developer 1	Build REST Controller — GET /all paginated, GET /{id} with 404, POST /create with @Valid, all correct status codes		Not Started

		Java	Java Developer 2	Build REST — PUT /{id}, DELETE (soft), GET /search?q=. Build create/edit form with validation in React	Not Started
		AI	AI Developer 1	Build POST /recommend — 3 recommendations as JSON array, each with action_type, description, priority	Not Started
		AI	AI Developer 2	Write AiServiceClient.java — RestTemplate calls to all Flask endpoints, 10s timeout, null return on error	Not Started
Day 5	Fri Apr 18	— Week 1 —			
		Java	Java Developer 1	JWT auth — JwtUtil, JwtAuthFilter, SecurityConfig. Build AuthController: /login, /register, /refresh	Not Started
		Java	Java Developer 2	React login page — AuthContext, ProtectedRoute, connect list page to real GET /all API with pagination	Not Started
		AI	AI Developer 1	Integrate AiServiceClient into Service — call AI on create @Async, attach result, handle null gracefully	Not Started
		AI	AI Developer 2	Week 1 security test — empty input, SQL injection, prompt injection on all endpoints. Document in SECURITY.md	Not Started
Day 6	Mon Apr 21	— Week 2 —			
		Java	Java Developer 1	Redis caching — @Cacheable on GETs (10 min TTL), @CacheEvict on writes. RBAC with @PreAuthorize on endpoints	Not Started
		Java	Java Developer 2	Dashboard — 4 KPI cards from /stats, Recharts chart. Detail page with score badge, Edit/Delete buttons	Not Started
		AI	AI Developer 1	Build POST /generate-report — structured JSON with title, summary, overview, key items, recommendations	Not Started
		AI	AI Developer 2	Prompt tuning — 10 real inputs per prompt, score accuracy, rewrite any scoring below 7/10	Not Started
Day 7	Tue Apr 22	— Week 2 —			
		Java	Java Developer 1	Email notifications — JavaMailSender, Thymeleaf templates, @Scheduled daily reminder and deadline alert	Not Started
		Java	Java Developer 2	Search/filter bar — debounced search, status dropdown, date range. Audit logging via Spring AOP on CUD methods	Not Started
		AI	AI Developer 1	Build GET /health — model, avg response time, uptime. Add Redis AI cache — SHA256 key, 15 min TTL	Not Started
		AI	AI Developer 2	Run OWASP ZAP scan — export report, fix all Critical findings today, plan Medium fixes	Not Started
Day 8	Wed Apr 23	— Week 2 —			
		Java	Java Developer 1	@ControllerAdvice — 404/400/500 consistent JSON. 10 JUnit 5 unit tests for Service with Mockito	Not Started
		Java	Java Developer 2	MockMvc tests — all endpoints, correct status codes. AI panel in UI — loading spinner, formatted response card	Not Started
		AI	AI Developer 1	Fix all ZAP findings — add security headers, re-scan confirms zero Critical/High remaining	Not Started

		AI	AI Developer 2	Write 8 pytest unit tests — mock Groq, test each endpoint format, error handling, injection rejection	Not Started
Day 9	Thu Apr 24	— Week 2 —			
		Java	Java Developer 1	docker-compose.yml — all 5 services, healthchecks, test with docker-compose up --build	Not Started
		Java	Java Developer 2	Swagger/OpenAPI on all endpoints. Add GET /export CSV. File upload POST /upload with type and size validation	Not Started
		AI	AI Developer 1	Final AI optimisation — all endpoints under 2s average, fallback template on Groq error ({is_fallback: true})	Not Started
		AI	AI Developer 2	Week 2 security sign-off — JWT, rate limit, injection all verified. PII audit confirms no personal data in prompts	Not Started
Day 10	Fri Apr 25	— Week 2 —			
		Java	Java Developer 1	Full integration test — docker-compose down -v then up, walk every feature, fix all bugs found	Not Started
		Java	Java Developer 2	Data seeder — 15 realistic demo records on startup. Analytics page with Recharts charts and period selector	Not Started
		AI	AI Developer 1	Write ai-service/README.md — setup, env vars, run instructions, API reference with examples for all 3 endpoints	Not Started
		AI	AI Developer 2	Week 2 AI quality review — 10 fresh inputs per endpoint, score accuracy (target avg >= 4/5). Fix failing prompts	Not Started
Day 11	Mon Apr 28	— Week 3 —			
		Java	Java Developer 1	Fix all Week 1-2 bugs. Additional tests to reach 80% coverage. UX polish — skeletons, empty states, Error Boundary	Not Started
		Java	Java Developer 2	Responsive design — 375px/768px/1280px all working. Performance — EXPLAIN ANALYZE, add indexes, fix N+1	Not Started
		AI	AI Developer 1	Pre-load sentence-transformers at startup. Full ZAP active scan — fix all Critical/High findings	Not Started
		AI	AI Developer 2	Full E2E test — docker-compose up, all AI integrations working correctly in containerised environment	Not Started
Day 12	Tue Apr 29	— Week 3 —			
		Java	Java Developer 1	Data seeder upgrade to 30 records. UI brand polish — #1B4F8A primary, Arial font, 8px grid, 44px touch targets	Not Started
		Java	Java Developer 2	Team code review — no secrets, no TODO comments, standards enforced. Final merge, tag release/v1.0	Not Started
		AI	AI Developer 1	Seed ChromaDB with 10 domain knowledge documents. Run all prompts against 30 demo records — all outputs demo-ready	Not Started
		AI	AI Developer 2	Final SECURITY.md — executive summary, all threats, tests, findings fixed, residual risks, team sign-off	Not Started

Day 13	Wed Apr 30	— Week 3 —			
		Java	Java Developer 1	Write README.md — overview, architecture diagram, prerequisites, setup steps, .env reference table	Not Started
		Java	Java Developer 2	Record 90-sec demo video — all key features with voice narration, save as demo_video.mp4	Not Started
		AI	AI Developer 1	Package AI — Dockerfile builds cleanly, requirements.txt exact versions, .env.example complete	Not Started
		AI	AI Developer 2	Final security checklist — all 4 members sign off, commit final SECURITY.md	Not Started
Day 14	Thu May 1	— Week 3 —			
		Java	Java Developer 1	Demo Rehearsal 1 — full team, stopwatch, complete 6-min presentation, note all issues	Not Started
		Java	Java Developer 2	Fix all rehearsal issues. Prepare 3 slides — problem, architecture, demo flow (1 min total for slides)	Not Started
		AI	AI Developer 1	AI dry run on demo machine — all 3 endpoints live, record response times, prepare backup screenshots	Not Started
		AI	AI Developer 2	Write AI demo script — exact inputs, expected outputs, 60-sec tech explanation for non-technical panel	Not Started
Day 15	Fri May 2	— Week 3 —			
		Java	Java Developer 1	Submit GitHub — public, no secrets, one commit per day, share link with mentor	Not Started
		Java	Java Developer 2	Fresh machine test — clone, create .env, docker-compose up --build, full stack healthy	Not Started
		AI	AI Developer 1	Submit AI service GitHub link — Dockerfile builds, README complete, share link with mentor	Not Started
		AI	AI Developer 2	Create 1-page AI summary card — 3 endpoints, tech stack, GitHub link. Print 2 copies for Demo Day	Not Started
Day 16	Mon May 5	— Week 4 —			
		Java	Java Developer 1	Final bug fix — zero P1/P2 remaining. 5 portfolio screenshots at 1920x1080 for each key screen	Not Started
		Java	Java Developer 2	Prepare 3 demo scenarios — exact steps, expected output, talking points. Confirm all work on live system	Not Started
		AI	AI Developer 1	Final performance verify — all endpoints within targets, cache working, fallback working	Not Started
		AI	AI Developer 2	Print AI talking points card — Groq, prompts explained in plain English. Prepare security talking points	Not Started
Day 17	Tue May 6	— Week 4 —			
		Java	Java Developer 1	Rehearsal 2 — full team, 6 min target, every member answers 5 key questions without notes	Not Started

		Java	Java Developer 2	Final hotfix window — P1/P2 only, re-test after each fix	Not Started
		AI	AI Developer 1	Groq API check — key active, credits sufficient, test all 3 endpoints confirmed working	Not Started
		AI	AI Developer 2	docker-compose down -v then up — fresh seeded state, all records visible, all scenarios confirmed	Not Started
Day 18	Wed May 7	— Week 4 —			
		Java	Java Developer 1	Final confidence check — each member presents their 90-sec section solo, resolve all remaining gaps	Not Started
		Java	Java Developer 2	Submit final SECURITY.md — professional, complete, accurate. Print 1 copy for Demo Day	Not Started
		AI	AI Developer 1	DEMO — Open: problem (1 sentence), architecture, launch live tool. Create record, watch AI respond	Not Started
		AI	AI Developer 2	DEMO — AI Recommend, Generate Report. Explain Flask + Groq in 60 seconds. Show /health endpoint	Not Started
Day 19	Thu May 8	— Week 4 —			
		Java	Java Developer 1	DEMO — Dashboard KPIs, search/filter, CSV export, audit log, email notification, responsive view	Not Started
		Java	Java Developer 2	DEMO — Security: 401 without token, 400 on injection, reference SECURITY.md, state findings fixed	Not Started
		AI	AI Developer 1	Post-demo — push final code, confirm all submissions complete, celebrate the sprint	Not Started
		AI	AI Developer 2	Post-demo — document lessons learned, note features for future sprints, feedback to mentor	Not Started
Day 20	Fri May 9	— Demo —			
		Java	Java Developer 1	DEMO — Open: problem statement, architecture slide, launch live tool, CRUD demonstration	Not Started
		Java	Java Developer 2	DEMO — Show UI features: dashboard, search/filter, CSV export, responsive design on mobile	Not Started
		AI	AI Developer 1	DEMO — Live AI: Describe, Recommend, Generate Report. Explain what AI is doing at each step	Not Started
		AI	AI Developer 2	DEMO — Explain tech stack. Security: 401 demo, injection rejection, SECURITY.md reference	Not Started

6. GITHUB — Push Code Every Working Day

Push your code at the end of every working day. Your mentor will share the repository details before Day 1.

Step	Action	Instructions
1	Create Account	Go to github.com, click Sign Up. Use your personal email. Choose a professional username.
2	Create Repository	Your mentor will share the exact repository name and URL. Set Visibility to Public. Add README.
3	Install Git	Download from git-scm.com. In terminal run: git --version to confirm version 2.x or higher.
4	Configure Git	Run: git config --global user.name 'Your Name' and git config --global user.email 'you@email.com'
5	Clone Repo	Run: git clone [repository URL shared by mentor] then cd into the project folder
6	Add .gitignore	On Day 1, before any commit: create .gitignore and add: .env target/ node_modules/ __pycache__/ chroma_data/
7	Daily Push	End of every working day: git add . then git commit -m 'Day X — what you did' then git push origin main
8	Verify Online	After every push: open the repository URL in your browser and confirm the latest files are visible.

Commit message format every day: Day X — brief description of what you completed

Example: Day 3 — Implemented Service layer with validation and custom exceptions

7. DEMO DAY CHECKLIST

Every item must be complete before your 6-minute Demo Day presentation on 9 May 2026.

Deliverable	What Must Be Submitted or Demonstrated	Done?
GitHub Repo	Public repo (link from mentor). At least 1 commit per working day. No secrets committed.	☐
README.md	Complete setup instructions. ASCII architecture diagram. Prerequisites. Step-by-step setup.	☐
docker-compose	All services defined. Full stack starts with docker-compose up. Tested on a clean machine.	☐
Flyway	All schema changes in numbered files (V1__, V2__, ...). No manual SQL outside migrations.	☐
JUnit Tests	Minimum 10 tests, all passing. JaCoCo coverage $\geq 80\%$.	☐
Pytest Tests	Minimum 8 tests, all passing. Groq API mocked — tests run without live network access.	☐
Swagger UI	All endpoints documented at /swagger-ui.html with request/response examples.	☐
AI Service	Flask app with working endpoints. Dockerfile builds. README covers setup.	☐
SECURITY.md	Threat model, tests conducted, findings fixed, residual risks, team sign-off.	☐
Demo Data	30 realistic seeded records on docker-compose up, covering all statuses and ranges.	☐
Demo Video	demo_video.mp4: 90-120 seconds, all key features shown with clear voice narration.	☐

8. DEMO DAY — Friday 9 May 2026

You have **6 minutes**. The tool must be running live on your laptop. No slides-only presentations. Real features, real data, live AI calls.

Segment	Time	What to Do and Say
Opening	1 min	State the problem in one sentence. Show architecture slide. Say 'Let me show you the live tool.'
CRUD Demo	2 min	Create a record live, watch AI description appear. Show audit log. Search and filter demo.
AI Features	1.5 min	Click AI Recommend (read aloud). Generate Report. Explain what the AI is doing.
UI + Security	1 min	Dashboard. CSV export. API call without JWT (show 401). Reference SECURITY.md.
Q&A;	0.5 min	Every member answers: What does it do? What AI? What security measures?

9. COMMON MISTAKES — Read This Now

These 8 mistakes are made every sprint. Avoiding them saves days of wasted work.

Mistake	Why It Happens and How to Avoid It
[X] Hardcoding secrets in config files	Use <code>\${ENV_VAR}</code> in <code>application.yml</code> and <code>os.getenv()</code> in Python. Actual values go only in <code>.env</code> (which is in <code>.gitignore</code>). If you commit a secret to GitHub, rotate it immediately — deleting the file is not enough.
[X] Committing <code>.env</code> to GitHub	Add <code>.env</code> to <code>.gitignore</code> on Day 1 before any commit. Run <code>git status</code> before every git commit to verify. GitHub scans for exposed keys and will warn you, but by then it is already public.
[X] Not wrapping Groq calls in try-except	The Groq free tier has rate limits and occasional outages. Every call must be in a try-except block. On failure: log the error, return the fallback template with <code>{is_fallback: true}</code> . Never return HTTP 500 because the AI is unavailable.
[X] Not setting up docker-compose until Week 3	Set it up on Day 5. The most common Demo Day failure is 'works on my laptop but not the demo machine.' Test the full docker-compose stack every day from Day 5 onwards. Discover integration problems early.
[X] Adding features in Week 4	Week 4 is for demo preparation only. Feature freeze after Day 15. Every hour building features in Week 4 is an hour not spent making the demo flawless. New ideas go in GitHub Issues labelled post-sprint.
[X] Empty database on Demo Day	Implement the DataLoader seeder on Day 14. On Demo Day: docker-compose down -v then docker-compose up to reset to a clean seeded state. An empty list on Demo Day undermines the entire presentation.
[X] Not reading Groq docs before writing AI code	Spend 30 minutes at console.groq.com/docs before Day 2. Key parameters: model name, temperature (0.3 for factual, 0.7 for creative), <code>max_tokens</code> , and the messages array structure. Misunderstanding temperature alone can waste an entire day.
[X] Skipping the demo dry run	Every team that skips the rehearsal on Day 17 regrets it. Problems are always found in rehearsal that you never noticed before. Do two full rehearsals (Day 17 and Day 19). Use a stopwatch. The time limit is firm.